

Hungarian Method

1 Purpose

Hungarian Method

The **Hungarian Method** solves the square assignment problem in polynomial time. It finds a minimum-cost one-to-one assignment between rows and columns.

For an $n \times n$ cost matrix $C = [c_{ij}]$, choose exactly one entry from each row and exactly one entry from each column so that the total cost is minimum.

Assignment Objective

$$\min_{\pi} \sum_{i=1}^n c_{i,\pi(i)}$$

where π is a permutation of the job columns.

2 Algorithm Steps

1. Subtract the minimum value of each row from every entry in that row.
2. Subtract the minimum value of each column from every entry in that column.
3. Cover all zeros using the minimum number of horizontal and vertical lines.
4. If the number of covering lines is n , choose independent zeros.
5. If fewer than n lines are needed, adjust the matrix and repeat.
6. Compute the final cost from the original cost matrix.

Independent Zeros

Independent zeros are zeros chosen so that no two selected zeros lie in the same row or the same column.

3 Original Matrix

Person	J_1	J_2	J_3	J_4	row min
A	6	5	7	7	5
B	12	8	9	3	3
C	7	6	9	10	6
D	6	3	7	8	3

The row minima are

5, 3, 6, 3.

4 Row Reduction

Subtract each row minimum from every entry of that row.

$$\begin{aligned}A &: (6, 5, 7, 7) - 5 = (1, 0, 2, 2), \\B &: (12, 8, 9, 3) - 3 = (9, 5, 6, 0), \\C &: (7, 6, 9, 10) - 6 = (1, 0, 3, 4), \\D &: (6, 3, 7, 8) - 3 = (3, 0, 4, 5).\end{aligned}$$

After row reduction:

Person	J_1	J_2	J_3	J_4
A	1	0	2	2
B	9	5	6	0
C	1	0	3	4
D	3	0	4	5

5 Column Reduction

Column minima after row reduction are

$$\min(J_1) = 1, \quad \min(J_2) = 0, \quad \min(J_3) = 2, \quad \min(J_4) = 0.$$

Subtract these values column by column.

Person	J_1	J_2	J_3	J_4
A	0	0	0	2
B	8	5	4	0
C	0	0	1	4
D	2	0	2	5

6 Zero Structure

After row and column reduction, zeros represent assignments that are cheapest relative to their row and column reductions. The goal is not to choose every zero. The goal is to choose n independent zeros.

Independent-Zero Requirement

For a 4×4 matrix, a complete assignment needs exactly four selected zeros:

one zero in each row and one zero in each column.

How to Choose Zeros Safely

Do not choose zeros randomly. A useful rule is to select forced zeros first.

1. Find a row with only one available zero and select that zero.
2. Cross out all other zeros in the selected zero's column.

- Find a column with only one available zero and select that zero.
- Cross out all other zeros in the selected zero's row.
- Repeat until each row and each column has exactly one selected zero.

If There Is a Tie

If several choices are possible, guessing may lead to a dead end. The safest interpretation is to treat zeros as a matching problem: match every row to a different zero-column. If a full matching is possible, the selected zeros give a valid optimal assignment.

For this matrix, the systematic selection is:

Step	Reason
Choose $B \rightarrow J_4$	Row B has only one zero.
Choose $D \rightarrow J_2$	Row D has only one zero.
Choose $C \rightarrow J_1$	After J_2 is used, row C 's safe available zero is J_1 .
Choose $A \rightarrow J_3$	Jobs J_1, J_2, J_4 are already used, so A takes J_3 .

Thus, one valid independent-zero selection is:

$$A \rightarrow J_3, \quad B \rightarrow J_4, \quad C \rightarrow J_1, \quad D \rightarrow J_2.$$

	J_1	J_2	J_3	J_4
A	0	0	0	2
B	8	5	4	0
C	0	0	1	4
D	2	0	2	5

Why This Selection Works

Check the selected rows:

$$A, \quad B, \quad C, \quad D$$

all appear exactly once.

Check the selected columns:

$$J_3, \quad J_4, \quad J_1, \quad J_2$$

also appear exactly once. Therefore, the selected zeros are independent.

Key Point

Other zeros may remain unselected. That is allowed. A valid assignment only needs one selected zero per row and one selected zero per column.

7 Cost from Original Matrix

The reduced matrix is used to find the assignment pattern. The final objective value must be computed from the original matrix.

Assignment	Original cost
$A \rightarrow J_3$	7
$B \rightarrow J_4$	3
$C \rightarrow J_1$	7
$D \rightarrow J_2$	3

$$\text{Minimum total cost} = 7 + 3 + 7 + 3 = 20.$$

8 Common Mistake: Feasible Does Not Mean Optimal

The following assignment is valid:

$$A \rightarrow J_1, \quad B \rightarrow J_4, \quad C \rightarrow J_3, \quad D \rightarrow J_2$$

It is feasible because every person gets exactly one job and no job is repeated.

However, feasibility alone does not prove optimality. Its original cost is

$$6 + 3 + 9 + 3 = 21.$$

Important Distinction

The Hungarian Method is not looking for just any feasible assignment. It is looking for the feasible assignment with minimum total cost.

The independent-zero assignment found above has cost

$$20.$$

Since

$$20 < 21,$$

the cost 21 assignment is valid, but it is not optimal.

9 Why Row and Column Reductions Work

Subtracting a constant from every entry of a row decreases the cost of every complete assignment by the same amount, because every complete assignment chooses exactly one entry from that row.

Similarly, subtracting a constant from every entry of a column decreases the cost of every complete assignment by the same amount, because every complete assignment chooses exactly one entry from that column.

Key Invariance

Row and column reductions change all complete assignment costs by the same total constant, so they do not change which assignment is optimal.

10 Complexity and Class

The Hungarian Method runs in polynomial time:

$$\mathcal{O}(n^3).$$

Therefore, the square assignment problem belongs to class P . The standard assignment problem is not NP-complete.

11 Final Review

Hungarian Method Essentials

Row reduction creates at least one zero in each row.
Column reduction creates at least one zero in each useful column.
Choose independent zeros systematically, not randomly.
Equivalently, find a full matching among zero positions.
Final cost is computed using the original matrix.
For this matrix, the optimum is 20.